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Transformative trends and challenges: Navigating disruptive innovation in higher education through bibliometric insights

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ABSTRACT

Disruptive innovation, provides a framework for examining how established industries, such as higher education, address challenges and pursue sustainable development. This paper reviews the evolution, current state, and future directions of disruptive innovation in higher education, highlighting key trends and areas for further research. A bibliometric analysis of 546 peer-reviewed publications (2001–2024) from the Web of Science was conducted using keyword analysis, clustering, and network visualisation. Core terms identified include higher education, COVID-19, disruptive innovation, education, and teacher education. Seven thematic clusters emerged: inclusive education, tech-pedagogy advancement, distance assessment, crisis education, educational AI, digital innovation, and transformative pedagogy. This study maps the intellectual landscape and identifies future research priorities to inform educational practices and policies. The dynamic environment of higher education offers opportunities for rethinking traditional models and advancing innovations in pedagogy and assessment, guiding policymakers, educators, and researchers towards fostering sustainable educational transformation.

KEYWORDS

Disruptive innovation; higher education; pedagogical innovation; educational technology; bibliometric analysis

Introduction

Disruptive innovation provides a framework for understanding why even well-established organisations can struggle despite their strengths (Christensen, 1997). It explains how companies focused on mainstream market innovations can be overtaken by new entrants offering disruptive technologies. These innovations may initially perform poorly in areas that mainstream customers value but introduce new features that attract niche or emerging markets. Over time, as these innovations improve, they begin to appeal to the broader market, ultimately displacing established technologies and contributing to the failure of incumbent firms. Although originally applied to business contexts, this framework is increasingly relevant for higher education institutions, which face similar competitive pressures and challenges as businesses (Taylor, 2017). By viewing universities as organisations navigating competitive environments, we can gain insights into how they might embrace disruptive innovation to adapt to technological changes and evolving student needs. Disruptive innovation has become a prominent topic in both academic

and business fields, playing a key role in research related to strategic management and innovation (Li et al., 2018). Recent work, such as Christensen et al. (2018), continues to advance the theory, demonstrating its ongoing relevance. Understanding disruptive innovation is essential for both businesses and higher education institutions aiming to foster sustainable growth and remain competitive.

The COVID-19 pandemic has intensified the focus on disruptive innovation, especially in education. The pandemic severely challenged traditional educational methods, necessitating innovative solutions to adapt to the new normal (Casper et al., 2022). Consequently, the role of disruptive innovation in reshaping educational practices has gained prominence. Examining disruptive innovation in higher education helps institutions understand emerging trends and adapt their strategies to remain competitive (Danneels, 2004). Disruptive innovations can democratise education by enhancing accessibility and affordability. Online learning platforms and Massive Open Online Courses (MOOCs) have disrupted traditional higher education models by providing flexible learning options to a broader audience (Kursun, 2016; Pilli et al., 2018). Recent studies explore the influence of artificial intelligence (AI) chatbots, specifically GPT (Generative Pre-trained Transformer), on the assessment practices of university teachers (Farazouli et al., 2024). They highlight the opportunities and challenges these chatbots present in education (Adeshola & Adepoju, 2024), underscoring the need for further research in this area (García-Peñalvo, 2023; Skavronskaya et al., 2023).

While previous research on disruptive innovation has examined various aspects, including frameworks and driving factors, many reviews rely on subjective qualitative methods, which may be vulnerable to bias (Shang et al., 2020; Tranfield et al., 2003). To address this, bibliometric analysis provides an objective, systematic approach to understanding the field's structure, research trends, and emerging concepts (Aria & Cuccurullo, 2017; Kraus et al., 2024). Unlike other methods, bibliometrics offers a macro-level view of the research landscape, revealing dynamics and offering objectivity (Öztürk, 2021; Zupic & Čater, 2015). Additionally, while disruptive innovation remains complex and underexplored, particularly in education (Christensen et al., 2013; Danneels, 2004), bibliometric analysis offers valuable insights.

This study highlights the connection between disruptive innovation and higher education, focusing on global transformations by identifying the evolution of research themes, emerging areas, shifts in focus, and development of research networks in the education industry over recent decades. Using bibliometric analysis of 546 peer-reviewed publications (2001–2024), it examines progress in disruptive innovation and technological advancements in education through metrics like keywords and co-occurrences. Addressing gaps in evaluation and discussion of resulting maps in current bibliometric studies (Block & Fisch, 2020), the study analyzes the conceptual and intellectual structure of these clusters and provides recommendations for future research.

An overview of disruptive innovation in higher education

Disruptive innovation explains how emerging technologies or business models displace established industries (Christensen, 1997). In higher education, this concept has driven significant changes in teaching methodologies, learning strategies, and accessibility, challenging traditional business models to improve affordability, flexibility, and

personalisation. From the 1990s onward, technological advancements and new approaches have reshaped the educational landscape, aligning with broader disruptions seen in other industries.

The 1990s saw the emergence of online education facilitated by learning management systems (LMS) like Blackboard and WebCT, enabling universities to offer courses online and expand access to education (Allen & Seaman, 2017). Concurrently, for-profit online universities such as the University of Phoenix gained popularity, catering to adult learners and working professionals, and disrupting traditional higher education (Means et al., 2016).

In the 2000s, the growth of open educational resources (OER), exemplified by initiatives like Massachusetts Institute of Technology (MIT)'s OpenCourseWare project, challenged traditional textbook publishing by providing free access to high-quality educational materials (Iiyoshi & Kumar, 2008). Later in the decade, the rise of MOOCs through platforms like Coursera, edX, and Khan Academy made high-quality courses available at little to no cost, drawing millions of global learners and further challenging conventional higher education models (Ho et al., 2015).

Parallel to these developments, the concept of blended learning gained momentum, integrating online and in-person instruction. The flipped classroom model allowed students to engage with instructional content online at their own pace, fostering flexibility while retaining face-to-face interactions (Picciano, 2017). The shift towards lifelong learning became apparent, leading to the popularity of micro credentials and shorter courses emphasising essential competencies like critical thinking and digital literacy (Borup et al., 2019).

Rabin et al. (2020) highlight the shift in higher education from traditional, hierarchical models to more open, flexible, and collaborative approaches. Traditional business models in higher education, such as tuition-based and degree-centric structures, are being disrupted by emerging models emphasising open access, online learning, and collaboration, with innovations like open educational resources (OER) and MOOCs offering more affordable, scalable alternatives (Greco et al., 2006; Orr et al., 2018). This disruption forces educational institutions to rethink their business strategies, moving towards more inclusive and adaptable models (Bates, 2015).

The transformative disruptions from the 1990s to the present have equipped educational institutions with valuable insights for navigating contemporary challenges, including those posed by global events like the COVID-19 pandemic. A comprehensive understanding of these disruptive innovations, gleaned from thorough reviews of scholarly articles and bibliometric analyses, offers invaluable guidance for enhancing the accessibility, affordability, and learner-centred nature of education while upholding or elevating educational quality.

The rise of social software and gamification is disrupting education, enhancing collaboration and assessment while challenging traditional models and raising concerns about privacy (Dawson, 2008; Murugesan, 2007). These changes highlight broader issues like the digital divide affecting marginalised communities (Molock & Parchem, 2022) and the risk of digital neocolonialism in education (Adam, 2019). Gamification, although promoting engagement, is still underexplored in terms of effectiveness (Deterding et al., 2011; Gee, 2003). As higher education evolves within the 4th industrial revolution, a strategic framework is essential for navigating these disruptions (Teng et al., 2019). This study uses Web

of Science data to provide tools for evaluating and integrating innovations, ensuring education adapts to both technological and societal needs.

Methodology

The research methodology used the Web of Science (WOS) database, known for its high-quality, comprehensive coverage of scholarly articles across disciplines (Clarivate, n.d.). The search targeted English-language, peer-reviewed journals using keywords such as 'disruptive innovation', 'disruptive education', 'disruptive technology', and 'higher education'. Disruptive innovation and disruptive technology often overlap, with disruptive technology playing a key role in disruptive innovation (Li et al., 2018). Both terms are included in the analysis, but for clarity, 'disruptive innovation' is used consistently throughout, except for terms derived from the analysis. Excluding publication types like brief communications and conference proceedings ensured focus and rigour. A total of 546 research papers published between 2001 and 2024 were analysed. A 2001 study on school health (Lurie et al., 2001) discussed innovation strategy but did not specifically address disruptive innovation. The concept of disruptive innovation in education was first explored in Friga et al. (2003) in their article, 'Changes in Graduate Management Education and New Business School Strategies for the 21st Century'. Their research focused on shifts in business models, particularly regarding primary markets, products, and partnerships within the education sector's demand and supply.

Data from the Web of Science (WOS) was saved in text format for analysis. VOSviewer, a software tool commonly used in bibliographic analysis (van Eck & Waltman, 2010), was used to construct visualisation maps using natural language processing (NLP) to extract key terms from titles and abstracts. VOSviewer helped identify relationships and trends within the data, creating clear visual representations of complex bibliometric information. This facilitated the understanding of important concepts, research fronts, and collaboration patterns within scientific fields.

Results and discussion

Keywords analysis

In this study, keyword co-occurrence analysis is used to explore publication content, based on the premise that relationships between terms can highlight specific areas of study (López-Fernández et al., 2016). The goal of this analysis is to identify clusters or key research themes. Author keywords, rather than Keywords Plus, were chosen from the Web of Science (WoS) database, as they more accurately reflect an article's content. Author keywords represent terms selected by the authors to best describe their work, while Keywords Plus are terms added by the publisher based on article analysis (Zhang et al., 2016).

Figure 1 presents a network diagram of key co-occurrences, where keywords appearing at least five times were retained. This resulted in 43 keywords out of a total of 1760. The five most frequent keywords identified are 'higher education' (56 occurrences), 'COVID-19' (25 occurrences), 'disruptive innovation' (24 occurrences), 'education' (22 occurrences), and 'teacher education' (16 occurrences). These terms highlight the central

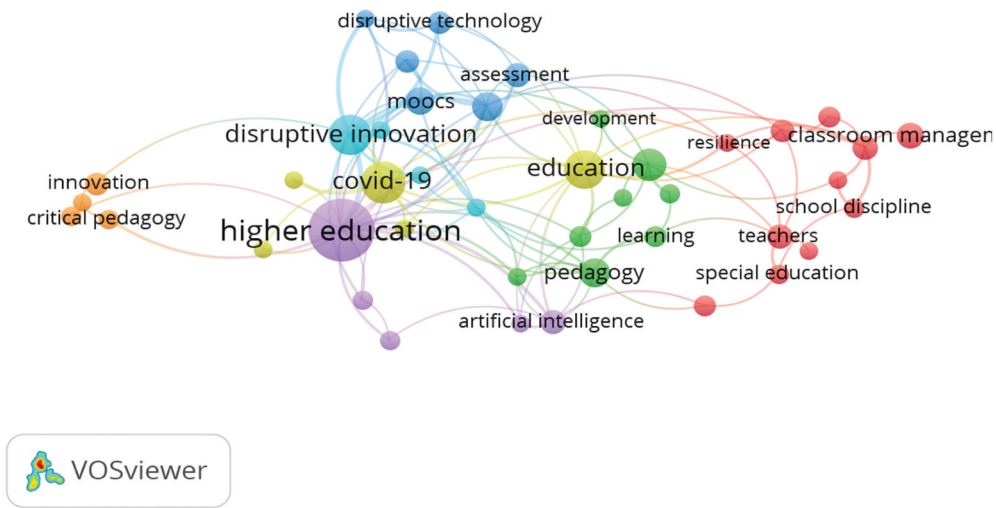


Figure 1. Network diagram of key co-occurrences. Source: Own elaboration.

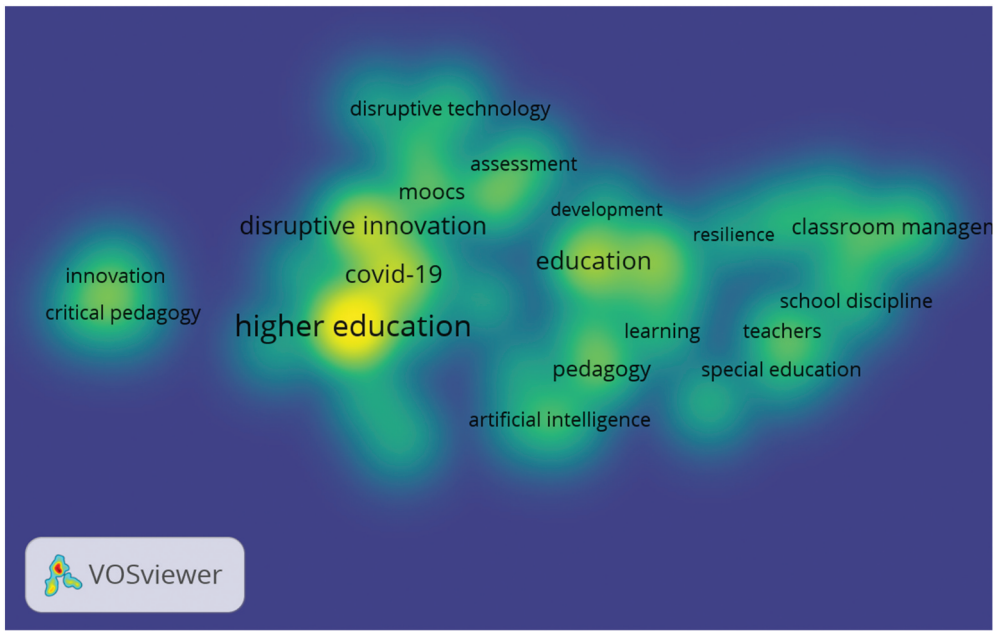


Figure 2. Density Visualization of key co-occurrences. Source: Own elaboration.

themes and areas of focus in the research, underscoring key trends in the field, particularly in relation to higher education.

Figure 2 presents a density visualisation of key co-occurrences, mapping the strength of relationships between frequently occurring keywords. Areas with higher density represent stronger associations, revealing how terms like ‘higher education’ and ‘disruptive innovation’ are closely linked. This visualisation highlights key research themes and their

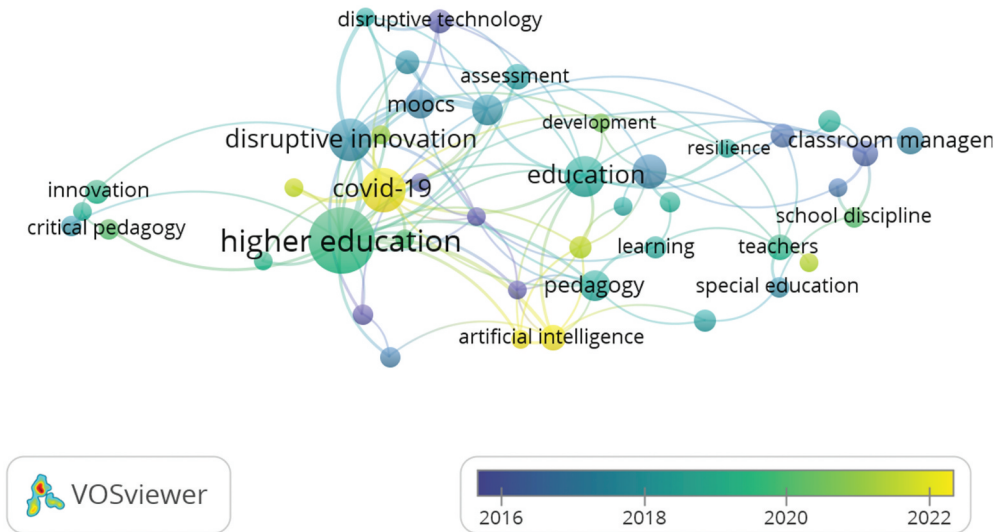


Figure 3. Overlay visualisation of key co-occurrences. Source: Own elaboration.

interconnectedness, with denser regions indicating more frequent and significant co-occurrences in the dataset, thus underscoring the centrality of certain concepts in the literature.

Figure 3 shows the overlay visualisation, depicting the temporal distribution of keywords across clusters. The colour gradient represents the average publication year: blue for earlier years, green for the middle period, and yellow for recent research. Keywords like 'COVID-19' and 'Artificial Intelligence' are shown in yellow, indicating their recent prominence and the evolving trends in disruptive innovation research in education. This highlights how certain topics have gained traction in response to current developments.

Thematic clusters

Figures 2, 3, and 4 illustrate the interconnected network, with all clusters linked by Cluster 5 (purple) at the core. Using VOSviewer, the Web of Science (WoS) data of the 43 references in Figure 2 were analysed to identify frequently occurring phrases in the keywords, titles, and abstracts. The abstracts and introductions of these references were also reviewed. Based on this analysis, the clusters were named. Table 1 lists the cluster names and their keywords in educational research on disruptive innovation.

- **Cluster 1: 'Inclusive Education'** promotes diverse, equitable, and socially just learning environments through inclusive teaching practices and early childhood education support. It addresses gender equality, resilience, and effective school discipline policies, and tackles racial disparities and special education needs (Beaman et al., 2006). It also examines biases in teacher attention, interventions to enhance novice teachers' classroom management, and the commercialisation of the Special Educational Needs sector, highlighting concerns about educational equity and inclusion (Tomlinson, 2021).

Table 1. Cluster names and their keywords in educational research.

Name of Cluster	Color	Cluster	Keywords
Inclusive Education	Red	Cluster 1 (11 items)	Classroom management, disruptive behaviour, early childhood, gender, inclusive education, race, resilience, school discipline, secondary education, special education, teachers
Tech-Pedagogy Advancement	Green	Cluster 2 (8 items)	Development, learning, pedagogy, professional development, teacher education, teaching, technology, university
Distance Assessment	Blue	Cluster 3 (6 items)	Assessment, disruptive technology, distance education, MOOCs, online learning, technology-enhanced learning
Crisis Education	Yellow	Cluster 4 (5 items)	COVID-19, education, pandemic, transformative learning, undergraduate
Educational AI	Purple	Cluster 5 (5 items)	Artificial intelligence, ChatGPT, disruptive technologies, higher education, open educational resources
Digital Innovation	Light Blue	Cluster 6 (4 items)	Disruptive innovation, e-learning, EdTech, online education
Transformative Pedagogy	Orange	Cluster 7 (4 items)	Critical pedagogy, innovation, social justice, transformation

- **Cluster 2: ‘Tech-Pedagogy Advancement’** emphasises the integration of innovative teaching methods, technology, and continuous professional development to enhance learning. It focuses on lifelong learning for educators and adapting pedagogical strategies to meet the digital age’s demands (Zembylas & Bekerman, 2008). Additionally, it explores how education can address historical trauma to promote empathy and understanding and examines factors affecting early-career mathematics teachers’ pedagogical competencies, including the impact of teacher education programmes, experience, and working conditions (König et al., 2015).
- **Cluster 3: ‘Distance Assessment’** explores the use of disruptive technologies to reshape assessment in online education (Al-Imarah & Shields, 2019). It highlights the role of MOOCs in broadening access to education while addressing quality assurance issues (Flavin & Quintero, 2018). Jones et al. (2010) examine the negative impact of social networking sites on students’ academic performance, noting that distractions can undermine focus and affect assessment outcomes in online learning environments. These findings underscore the challenges of maintaining academic integrity in distance education.
- **Cluster 4: ‘Crisis Education’** examines educational responses to crises like the COVID-19 pandemic, focusing on resilience and adaptability (Bashir et al., 2021). It highlights challenges faced by medical residents, such as disruptions in clinical training and heightened stress levels (Molock & Parchem, 2022). Additionally, it explores leadership during crises, emphasising community, communication, and compassion in managing such situations (Longmuir, 2023).
- **Cluster 5: ‘Educational AI’** examines the integration of AI, particularly GPT, in higher education, transforming teaching, learning, and administration. Studies highlight GPT’s potential in tourism education (Skavronskaya et al., 2023), its impact on assessment practices and challenges in replacing traditional exams (Farazouli et al., 2024), and concerns over privacy and overreliance (Adeshola & Adepoju, 2024).

Additionally, research explores educators' perceptions of GPT's disruptive potential and ethical concerns, including its effects on creativity (García-Peñalvo, 2023).

- **Cluster 6: 'Digital Innovation'** explores the transformative role of technology in education, focusing on disruptive innovation, EdTech, and online learning. Studies examine the evolving relationship between universities and EdTech post-MOOC, addressing benefits and challenges (Thomas & Nedeva, 2018), the impact of language in EdTech incubators (Ramiel, 2019), and the tension between innovation and equity in policy-making (Ramiel, 2021). The cluster also assesses the impact of e-learning, emphasising the need for alignment between technology, pedagogy, and context (Heilesen & Josephsen, 2008; Mentis, 2008).
- **Cluster 7: 'Transformative Pedagogy'** highlights the integration of technology and innovative pedagogy. Maida (2011) advocates for project-based learning (PBL) to foster collaboration and critical thinking. Farrow (2017) discusses open education's role in providing equitable, flexible learning. Hytten (2006) explores social justice in pedagogy, using digital tools to encourage diverse perspectives. Ličen and Jedlicka (2022) demonstrate technology's role in enhancing sustainability education in sport management, emphasising collaborative, experiential learning. These approaches prioritise technology, social justice, and student-centred practices.

Cluster structural observations

The interconnected network and cluster structure of disruptive innovation in higher education scholarship from 2001 to 2024 can be visualised as a concentric circle map, illustrating the relationships between key concepts. At the core is 'higher education', with branches extending to eight additional concepts: disruptive technology, COVID-19, classroom management, MOOCs, development, resilience, and critical pedagogy.

Visual representations, such as the density visualisation (Figure 1), network visualisation (Figure 2), and overlay visualisation (Figure 3), provide insights into keyword connections and their temporal distribution. A colour gradient from yellow (hotter) to blue (colder) indicates the strength of keyword presence and link strength, with darker blue representing earlier stages of research (e.g. classroom management) and yellow indicating recent areas of focus (e.g. artificial intelligence and COVID-19).

These visual aids facilitate the identification of research gaps in the literature that might otherwise be challenging to discern. One notable observation is the relative isolation of Cluster 1 (Inclusive Education) and Cluster 7 (Transformative Pedagogy), suggesting a disconnect between research in inclusive teaching practices and deeper explorations of innovative methods and technology integration. Research in these clusters emphasises critical pedagogy, social justice, and innovation in education, highlighting the need for empirical studies to explore their application in diverse educational settings.

Another observation reveals the emergence of a macro-cluster formed by crisis education (Cluster 4) and educational AI (Cluster 5), reflecting a growing interest in researching the educational response to crises such as COVID-19 and the integration of artificial intelligence in higher education. While existing studies shed light on specific challenges

and benefits, further research is needed to understand the broader societal impact and ethical dimensions of these technological advancements.

The complexity of tech-pedagogy advancement (Cluster 2), innovations in online assessment (Cluster 3), and digital innovation (Cluster 6) underscores the importance of aligning technology with pedagogy and context. Further exploration is required to understand the effectiveness and implications of integrating disruptive technologies into assessment practices, as well as their impact on educational outcomes and equity. Additionally, investigating the intersectionality between technology, pedagogy, and context in e-learning environments is essential for informing policy decisions and best practices in educational technology implementation.

Recommendations on how to fill research gaps

To address the identified research gaps, future studies should focus on several key areas to foster innovation and cohesion in education. First, bridging the gap between inclusive education and pedagogy is essential for advancing equity, social justice, and innovation. Research should explore how transformative pedagogy can manage disruptive behaviour within inclusive classrooms, incorporating student-led learning, flipped classrooms, and project-based approaches. Examining humanistic approaches – such as Koon’s (2023) integration of technology with empathy and AI’s ethical implications – may promote student well-being and critical thinking.

Further research should focus on teacher attention, professional development, and the commercialisation of special education services, along with understanding how student behaviour, contextual variations, and diverse stakeholder perspectives impact practices (Barber et al., 2013). Research aimed at improving novice teachers’ classroom management strategies could also contribute to more effective teaching.

Additionally, research should evaluate the effectiveness of digital tools in pedagogy, focusing on their impact on student outcomes, equity, and societal change. Darling-Hammond (2017) stresses that integrating technology into pedagogy fosters collaborative learning and critical thinking in student-centred environments. Investigating scalable and sustainable pedagogical models is crucial for refining the intersection of pedagogy and digital tools.

The role of disruptive technologies in online assessment also demands further exploration. Studies should investigate how MOOCs and social networking platforms influence assessment quality and integrity, while identifying strategies to mitigate challenges and optimise their use in diverse educational contexts.

Finally, understanding how technology, pedagogy, and context interact is vital for the effective implementation of digital innovations. Research should explore the ethical implications of using these technologies, ensuring their responsible integration into educational settings. By addressing these research gaps, education systems can better integrate technology with pedagogy, fostering innovation and creating more inclusive learning environments.

Conclusions, limitations and future directions

This study lays a significant foundation for future research on disruptive innovation in higher education, contributing both theoretically and methodologically. It enhances understanding of disruptive innovation's role in higher education and introduces a bibliometric approach to mapping complex educational disciplines (Caputo et al., 2021). The analysis identifies key clusters such as 'Inclusive Education', 'Tech-Pedagogy Advancement', and 'Educational AI', while also highlighting gaps in inclusive education and transformative pedagogy.

While there are limitations, including potential biases in the Web of Science database and the exclusion of some publication types, these can be addressed in future research to provide a more comprehensive view. Clustering yields useful insights, but human interpretation is crucial.

Future studies should explore the ethical, equitable, and pedagogical impacts of disruptive technologies (Bashir et al., 2021; Christensen, 1997). Additional analysis of the identified clusters can further illuminate emerging educational technologies and philosophies, including the impact of AI chatbots, such as GPT, on university assessment practices (Adeshola & Adepoju, 2024; Farazouli et al., 2024; García-Peñalvo, 2023; Skavronskaya et al., 2023).

The field of educational research is rapidly evolving, particularly as technology intersects with pedagogy. As educational systems adjust to a digital landscape, integrating technology into teaching and learning is central to improving educational outcomes and fostering transformative change. This dynamic environment presents opportunities for rethinking traditional educational models and advancing innovation in both pedagogy and assessment.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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